

Box 49.1 Strategies to Overcome Diuretic Braking

- 1. Restrict dietary salt to approximately 120 to 140 mmol daily to prevent postdiuretic salt retention.
- 2. Consider adding another class of diuretic.
- 3. Consider multiple daily dosing or a diuretic with prolonged action.
- 4. Do not stop diuretic therapy abruptly.
- 5. Prevent or reverse diuretic-induced metabolic alkalosis and hyponatremia.

Box_49_1

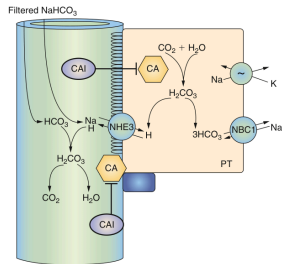


Fig. 49.2 Mechanisms of action of carbonic anhydrase inhibitors in the proximal tubule. The figure shows a functional model of proximal tubule cells; many transport proteins are omitted from the model for clarity. Inside the cell, carbonic anhydrase (CA) catalyzes the formation of H₂CO₃ from OH⁻ and CO₂. Bicarbonate leaves the cell via the sodium-bicarbonate transporter 1 (NBC1). A second pool of carbonic anhydrase is located in the brush border. This participates in disposing of the carbonic acid formed from filtered bicarbonate and secreted H⁺. Both pools of carbonic anhydrase are inhibited by acetazolamide and other carbonic anhydrase inhibitors (CAIs; see text for details).

Fig_49_2

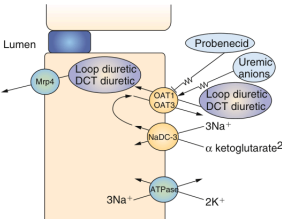


Fig. 49.5 Mechanisms of diuretic secretion by proximal tubule cells. Cell diagram of the S2 segment of the proximal tubule shows secretion of anionic diuretics including loop diuretics and distal convoluted tubule (DCT) diuretics. Peritubular uptake by an organic anion transporter (primarily OAT1, although OAT3 may play a smaller role) occurs in exchange for α -ketoglutarate, which is brought into the cell by the Na⁺-dependent cation transporter NaDC-3. Luminal secretion can occur via a voltage-dependent pathway or in exchange for luminal hydroxyl or urate. A portion of the luminal transport traverses multidrug resistance-associated protein 4 (Mrp4). ATPase, Adenosine triphosphatase.

Fig_49_5

Box 49.2 Common Causes of Diuretic Resistance

- Incorrect diagnosis (e.g., venous or lymphatic edema)
- Inappropriate NaCl or fluid intake
- Inadequate drug reaching tubule lumen in active form because of:
 - Noncompliance
 - Dose inadequate or too infrequent
 - Poor absorption (e.g., due to uncompensated HF)
 - Decreased renal blood flow (e.g., due to HF or cirrhosis of liver, or in older patients)
 - Decreased functional renal mass (e.g., due to AKI or CKD, or in older patients)
- Proteinuria (e.g., due to nephrotic syndrome)
- Inadequate renal response because of:
 - Low glomerular filtration rate (e.g., due to AKI, CKD)
 - Decreased effective extracellular fluid volume (e.g., due to edematous conditions)
- Activation of renin-angiotensin-aldosterone system (e.g., due to edematous conditions)
- Nephron adaptation (e.g., due to prolonged diuretic therapy)
- Nonsteroidal antiinflammatory drugs (e.g., indomethacin and aspirin)

AKI, Acute renal failure; CKD, chronic kidney disease; HF, heart failure.

Box_49_2

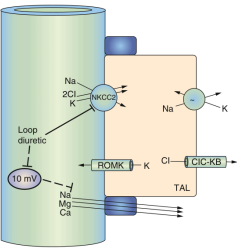


Fig. 49.3 Mechanisms of diuretic action along the loop of Henle. A model of thick ascending limb cells. Na⁺ and Cl⁻ are reabsorbed across the apical membrane via the loop diuretic-sensitive Na-K-2Cl cotransporter 2 (NKCC2). Loop diuretics bind to and block this pathway directly. Note that the transepithelial voltage along the thick ascending limb is oriented with the lumen positive relative to blood (circled value, given in millivolts [mV]). This transepithelial voltage drives a component of Na⁺ (and calcium and magnesium; see Fig. 49.4) reabsorption via the paracellular pathway. This component of Na⁺ absorption is also reduced by loop diuretics because they reduce the transepithelial voltage. ClC-KB, Chloride channel protein; ROMK, renal outer medullary K⁺ channel; TAL, thick ascending limb.

Fig_49_3

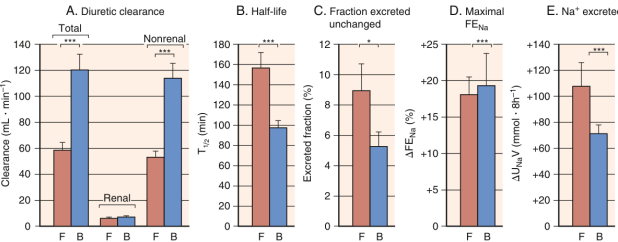


Fig. 49.6 (A–E) Comparison of the pharmacokinetics and dynamics of furosemide (F, 160 mg; metabolically inactivated in the kidney) and bumetanide (B, 4 mg; metabolically inactivated in the liver) in 10 subjects with chronic kidney disease (mean creatinine clearance, 12 ± 2 mL/min). Significance of difference: *, $P < .05$; **, $P < .01$; ***, $P < .005$. FE_{Na} , Fractional excretion of sodium; $T_{1/2}$, half-life; U_{Na} , urinary sodium. (Redrawn from data in Voelker JR, Cartwright-Brown D, Anderson S, et al. Comparison of loop diuretics in patients with chronic renal insufficiency. *Kidney Int*. 1987;32:572–578.)

Fig_49_6

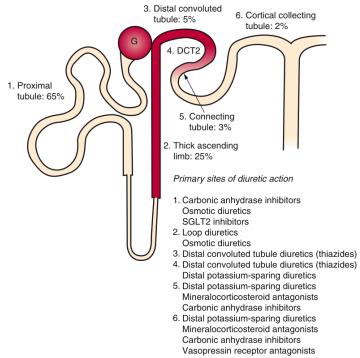


Fig. 49.1 Nephron diagram showing primary sites of diuretic action and approximate percentage of filtered sodium reabsorbed at each. DCT2, Late segment of distal convoluted tubule; G, glomerulus.

Fig_49_1

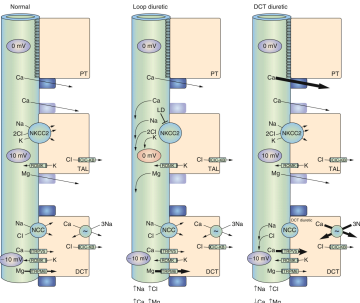


Fig. 49.4 Possible mechanisms of diuretic effects on calcium and magnesium excretion. Typical cells from the proximal tubule (PT), thick ascending limb (TAL), and distal convoluted tubule (DCT) are shown. Calcium reabsorption occurs along the DCT largely via a transient receptor potential channel (TRPV4) and magnesium reabsorption via a transient receptor potential channel (TRPV4). Transepithelial voltages (mV) are shown for each cell type. Normal conditions are at the left. Treatment with loop diuretics (LD) is shown in the middle; treatment with DCT diuretics is shown on the right. Loop diuretics reduce the magnitude of the lumen-positive transepithelial voltage, thereby reducing passive calcium and magnesium reabsorption via the paracellular pathway. Long-term treatment, especially with DCT diuretics, increases proximal Na⁺ and Ca²⁺ reabsorption, thus less calcium is delivered distally. Enhanced distal calcium absorption, driven by DCT diuretics, also occurs. Effects of DCT diuretics to increase magnesium secretion remain incompletely understood. ClC-KB, Chloride channel protein; NKCC2, Na-K-Cl cotransporter 2; ROMK, renal outer medullary K⁺ channel; L, increased; D, decreased.

Fig_49_4

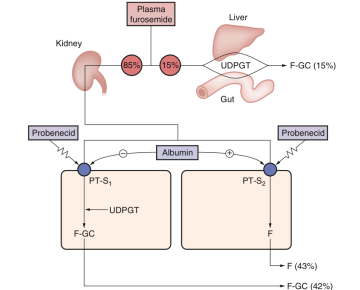


Fig. 49.7 Diagrammatic representation of the disposition of intravenous furosemide and the effects of hyponatremia or probenecid in normal or hyponatremic rabbits. After intravenous administration of furosemide, 15% is metabolized by uridine diphosphate glucosyl transferase (UDPGT) in the liver and gut to the inactive furosemide glucuronide (F-GC). Of the remainder, 85% is transported by the kidney. Some 42% is taken up in the S1 segment of the proximal tubule (PT-S1) and metabolized to the inactive glucuronide, and the remainder is taken up by the S2 segment (PT-S2) and secreted in active form into the lumen. Both uptake processes are inhibited by probenecid. The plasma albumin concentration facilitates uptake and secretion by PT-S2, but inhibits uptake and metabolism by PT-S1. (Modified from Pichette V, Gosselin D, du Souich P. The influence of moderate hyponatremia on the renal metabolism and dynamics of furosemide in the rabbit. *Br J Pharmacol*. 1990;119:885–890.)

Fig_49_7

49. Diuretics · assets 10-18 / 23

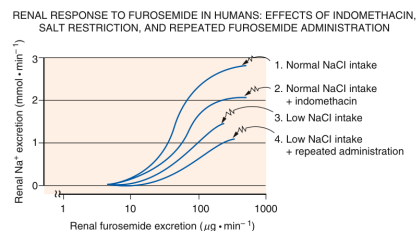


Fig. 49.8 Relationship between the excretion of Na⁺ and furosemide (log scale) following a bolus intravenous injection of 40 mg furosemide in normal subjects with a normal NaCl intake (1), with a normal NaCl intake after indomethacin (2), with a low Na⁺ intake (20 mmol/24 hours) (3), and for the third day of furosemide administration with a low Na⁺ intake (4). (Modified from Wilcox CS, Mitch WE, Kelly RA, et al. Response of the kidney to furosemide. *J Lab Clin Med*. 1983;102:450-458; and Chennavasin P, Seiwel R, Brater DC. Pharmacokinetic-dynamic analysis of the indomethacin-furosemide interaction in man. *J Pharmacol Exp Ther*. 1980;215:77-81.)

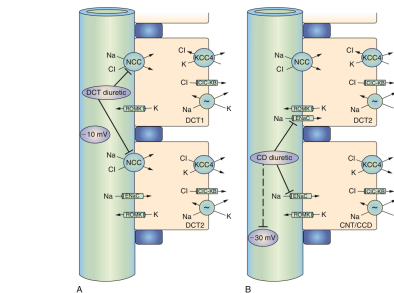


Fig. 9A5 Mechanisms of distal-evoked tubule (DET) and collecting duct (CD) diuresis. (A) Mechanism of Cl^- and H_2O diuresis. In rat, mouse, and humans, two types of DETs have been identified, referred to as DET1 and DET2. Na^+ and Cl^- are reabsorbed across the apical membrane of DET cells only via the thiazide-sensitive Na^+ / Cl^- cotransporter (NCC). This transport protein is also expressed by DET cells where Na^+ can also cross through the epithelial Na^+ channel (ENaC; see text for details). Thus, the trans-epithelial voltage along the DET1 is near zero. In the DET2, the luminal negative charge along the CD (B) promotes diuresis of Cl^- diuresis. The late distal convoluted tubule (LDC) and collecting duct (CD) are the sites of Cl^- secretion. The Cl^- secretion is driven by the Cl^- channel (CFTR) and the Cl^- cotransporter (KCC2) with a renal outer medullary K channel (ROMK). The trans-epithelial voltage is oriented with the lumen negative, relative to the interstitium (shown in the circled value), generating a favorable gradient for Cl^- secretion. Ducts that block the epithelial Na^+ channel reduce the voltage toward 0 mV (effect indicated by dashed line), thereby inhibiting K⁺ and H⁺ secretion. CCCa, Chloride channel protein, KCa, Ca^{2+} channel protein.

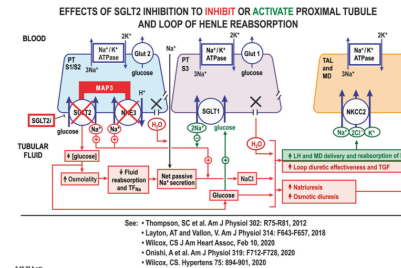


Fig. 4B.0 Effects of sodium-glucose-linked transport 2 inhibition on proximal tubule and loop of Henle reabsorption. A simplified functional model of the early (S1 and S2) and late (S3) segments of the proximal tubule and the thick ascending limb of loop of Henle. SGLT2 is linked functionally to the sodium hydrogen exchanger 3 and to the urate transporter (URAT1) by the microtubule-associated protein 3 (MAP3). Inhibition of SGLT2 blocks NHE3, thereby delivering more sodium from the proximal tubule while also reducing glucose reabsorption, thereby increasing tubular fluid osmolality to create an osmotic diuresis. More glucose is delivered to SGLT1 in the S3 segment that is activated to reabsorb sodium coupled in a 2:1 ratio with glucose that thereby offsets the natriuresis. The fall in tubular fluid sodium concentration promotes a proximal leak from the interstitium to the tubular fluid. Increased delivery of sodium chloride to the thick ascending limb (TAL) of loop of Henle enhances responsiveness to loop diuretics while increased delivery to the macula densa activates the tubuloglomerular feedback mechanism (TGF).

Fig 49 8

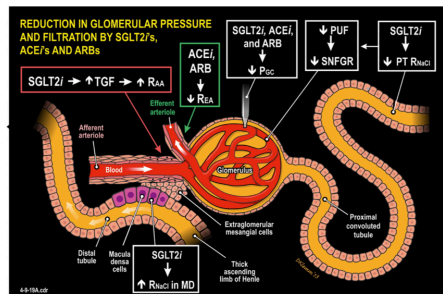


Fig 4B.11 A diagrammatic depiction of mechanisms that may reduce glomerular capillary pressure and reversal of glomerular hyperfiltration by a sodium-glucose-linked transport type 2 inhibitors (SGLT2is), angiotensin-converting enzyme inhibitors (ACEis) and angiotensin receptor blockers (ARBs). SGLT2i reduces reabsorption of sodium and chloride in the proximal tubule and increases their delivery and reabsorption at the macula densa segment. This activates the tubuloglomerular feedback (TGF) response to increase the resistance of the afferent arteriole and reduce the hydrostatic pressure in the glomerular capillaries. ACEis and ARBs lead to a greater reduction in resistance of the efferent than the afferent arteriole and reduce the hydrostatic pressure in glomerular capillaries by a different mechanism.

Fig 49 11

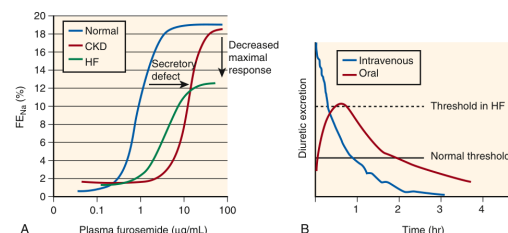


Fig. 4B.14 Dose-response curves for loop diuretics. (A) The fraction Na^+ excretion (FE_{Na}) as a function of plasma loop diuretic concentration. Compare with normal subjects, patients with chronic kidney disease (CKD) show a rightward shift in the curve owing to impaired distal secretion. The maximal response is preserved when expressed as FE_{Na} , but not when expressed as absolute Na^+ excretion. Patients with congestive heart failure (HF) demonstrate a rightward and downward shift, even when expressed as FE_{Na} , and thus are relatively diuretic-resistant. (B) Comparison of the response to intravenous and oral doses of loop diuretics. In a normal individual, an oral dose may be as effective as an intravenous dose because the time during which this individual is above the natriuretic threshold (indicated by the *Normal threshold* line) is approximately equal. If the natriuretic threshold increases (as indicated by the *CKD threshold* in HF line), then the oral dose may not provide a high enough plasma level to elicit natriuresis.

Fig 49 14

Fig 49 9

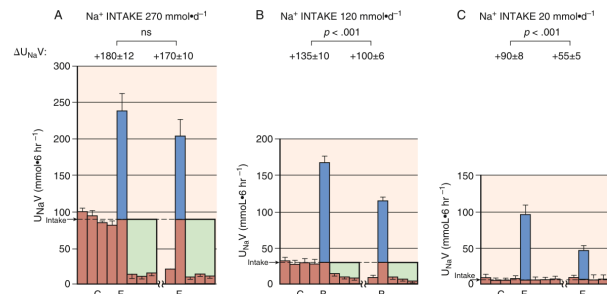


Fig. 49.12 Effects of dietary salt intake on the diuretic braking phenomenon. Renal Na⁺ excretion (mmol/6 hours) for 24 hours before and after intravenous (F₁) and third (F₃) daily doses of furosemide (A and C; 40 mg intravenously) and (B) subcutaneous (F₂, B₂) 40 mg intravenously in groups of 10 to 10 normal subjects equilibrated to fixed body Na⁺ intake (average level). Na⁺ intake (mmol/6 hours) shown by broken horizontal lines. Na⁺ balance is indicated by black bars and positive Na⁺ balance by light green areas. The mean $\pm$ standard error of the mean (SEM) values for diuretic-induced increases in Na⁺ excretion above baseline values (Δ U_{Na}⁺) for 6 hours after the administration of the diuretic are shown at the top. (Modified from Wilcox SS, Mitch WE, Kelly RA, et al. Response of the kidney to furosemide. *J Lab Clin Med*. 1983;102:450–458.)

Fig 49 12

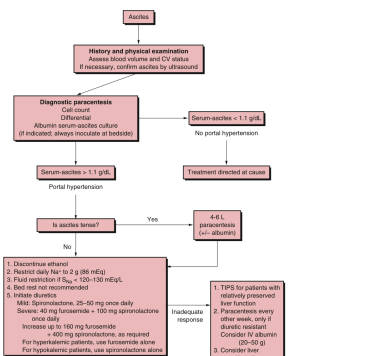


Fig. 49.15 Treatment algorithm for the management of fluid retention in patients with hepatic cirrhosis and ascites. CV, Cardiovascular; IV, intravenous; S_{Na} , serum Na⁺ concentration; TIPS, transjugular intrahepatic portosystemic shunt. [From Ellison DH, Wilcox CS. Diuretics: use in edema and the problem of resistance. In Brady HR, Wilcox CS, eds. *Therapy in Nephrology and Hypertension*, ed 2. London: Elsevier Science; 2003.]

Fig 49 15

Fig 49 10

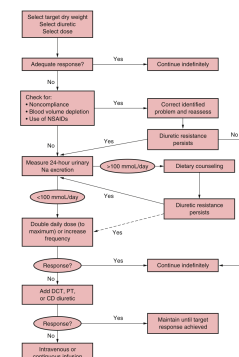


Fig. 49.13 Diagrammatic representation of an approach to the management of a patient with resistance to a loop diuretic. CD, Collecting duct diuretic (e.g., amiloride, furosemide, or spironolactone); DCT, distal convoluted tubule diuretic (e.g., thiazide); NSAIDs, nonsteroidal antiinflammatory drugs; PT, proximal tubule diuretic (e.g., acetazolamide). From Ellison DH, Wilcox CS. Diuretics: use in edema and the problem of resistance. In Brady HR, Wilcox CS, eds. *Therapy in Nephrology and Hypertension*, ed 2. London: Elsevier Science; 2003.

Fig 49 13

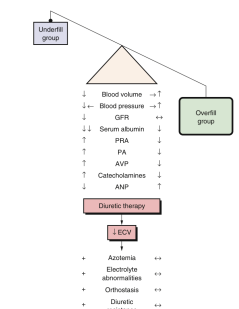


Fig. 49.16 Comparison of clinical and biochemical characteristics and responses in patients with nephrotic syndrome and underfill edema versus overfill edema. *AIP*, Atrial natriuretic peptide; *AVP*, arginine vasopressin; *IECV*, decrease in extracellular fluid volume; *GFR*, glomerular filtration rate; *PA*, plasma aldosterone; *PA*, plasma renin activity; *↑*, increase; *↓*, decrease; *↔*, no effect; *—*, potential side effect. (Modified from Schrier RW, Fassett AG. A critique of the overall hypothesis of sodium and water retention in the nephrotic syndrome. *Am J Med Sci*. 1998;326:1111-1117.)

Fig 49 16

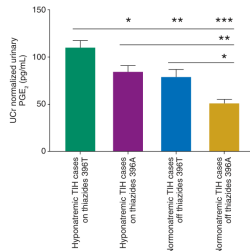


Fig. 49.17 Urinary prostaglandin E₂ (PGE₂) concentration in thiazide-induced hyponatremia (TH) cases, compared with normonatremic thiazide- and non-thiazide-treated controls, stratified according to their genotype at position 398 of the prostaglandin transporter gene, SLC02A1. TH cases who carry at least one copy of the risk allele (398T) have significantly elevated urinary PGE₂ levels relative to those homozygous for 398A, whereas no such effect was observed in normonatremic controls. Data are represented as mean ± SEM. *P < .05; **P < .01; ***P < .001. UC, Urinary creatinine. From Vaire J-S, Wan L, Charnak-Jaisak D, et al. Phenotypic and pharmacogenetic evaluation of patients with thiazide-induced hyponatremia. *J Clin Invest.* 2011;121(6):3367–3374.]

Fig_49_17

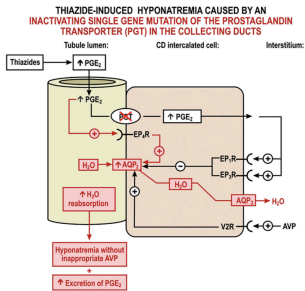
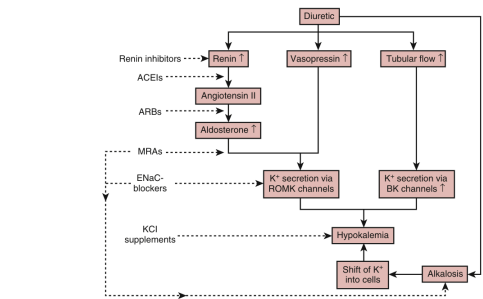
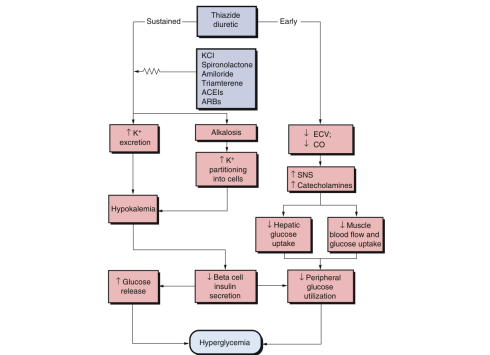


Fig. 49.18 Hypothesis for the role of SLC02A1 in contributing to thiazide-induced hyponatremia in individuals carrying the SLC02A1 398T variant. Thiazide diuretics increase renal synthesis and secretion of prostaglandin E₂ into the tubular fluid of the distal nephron. Tubular fluid PGE₂ is normally reabsorbed by the prostaglandin transporter in the distal nephron. PGE₂ activates the PG type 1 and 3 receptors (EP1 and EP3) on the basolateral membrane of the principal cells of the collecting duct (CD) that inhibit the insertion and activation of the aquaporin type 2 channel (AQP2) into the luminal membrane. Inactivation of the PGT limits PGE₂ from accessing the basolateral EP1 and EP3 receptors and increases its luminal concentration and activation of the luminal EP4 receptor that prompts the insertion and activity of AQP2. This results in increased reabsorption of free water and the development of hyponatremia coupled with increased excretion of PGE₂. AQP2, Aquaporin-2; PGE₂, prostaglandin E₂; PGT, phosphate glucuronyl transferase; TH, thiazide-induced hyponatremia. (Data from Palmer BF, Clegg DJ. Altered prostaglandin signaling as a cause of thiazide-induced hyponatremia. *Am J Kidney Dis.* 2018;71(6):769–771.)

Fig_49_18



Fig_49_19



Fig_49_20

Table 49.1 Ceiling Doses of Loop Diuretics (in mg)				
Condition	Furosemide		Bumetanide, IV or PO	Torsemide, IV or PO
	IV	PO		
Chronic renal insufficiency:				
• Moderate (GFR = 20–50 mL/min/1.73 m ²)	80–160	160	4–8	20–50
• Severe (GFR <20 mL/min/1.73 m ²)	160–200	240	8–10	50–100
Nephrotic syndrome with normal GFR	80–120	240	2–3	20–50
Cirrhosis with normal GFR	40–80	80–160	1	10–20
Heart failure with normal GFR	40–80	80–160	1–2	10–20

GFR, Glomerular filtration rate; IV, intravenous; PO, oral.
Data from Brater DC. Diuretic therapy. *N Engl J Med.* 1998;339:387–395.

Table_49_1